



Investor distraction and multi-dimensional financial narrative

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Abstract

This paper investigates how institutional investor distraction affects the assimilation of narrative content in the MD&A section of the 10-K filing. We introduce the Aggregate Attribute Index (AAI) and an alternative formulation (AltAAI), which capture linguistic features beyond tone to provide a broader measure of corporate narrative richness. Using machine learning and natural language processing, we analyze U.S. firms that follow a staggered reporting strategy, releasing quantitative results before full narrative disclosures. This design isolates the incremental effects of complex language when other portfolio events distract investors. We find that narrative complexity does not trigger short-term return responses but significantly affects stock prices over longer horizons. Complexity moderates how and when attention-constrained investors adjust prices. These effects are not captured by dictionary-based tone or readability metrics, underscoring the distinct role of multi-dimensional attributes in shaping delayed market reactions and price discovery.

Keywords Investor distraction · Institutional ownership · Textual analysis · Machine learning

JEL Classification G14 · G32 · M41

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1 Introduction

Investor attention has become a central concern for researchers (Hirshleifer and Teoh 2003; Sims 2003; Peng 2005; Peng and Xiong 2006; Hirshleifer et al. 2009; Banerjee and Kremer 2010; Hendershott et al. 2022) seeking to understand how financial markets incorporate complex corporate disclosures into stock prices. The expanding volume of textual information in the Management Discussion and Analysis (MD&A) section of the 10-K, the frequent release of supplementary filings, and the proliferation of qualitative information have all inhibited even professional investors' ability to absorb and interpret every piece of news in real-time. Research (Peng and Xiong 2006; DellaVigna and Pollet 2009; Li and Yu 2012; Chen et al. 2022) increasingly recognizes that attention scarcity may cause significant deviations from traditional notions of market efficiency. This in turn may affect returns and stock price predictability, prompting firms to adapt their disclosure strategies in anticipation of how and when distracted investors will respond (Bertomeu et al. 2023; Hendershott et al. 2022). Recent theoretical and empirical advances clarify these effects by examining how much firms disclose but raise the possibility that specific features of financial narratives may be misread or underappreciated when attention is constrained.

Although research (e.g., Tetlock 2007; Henry 1973; Tetlock et al. 2008; Feldman et al. 2010; Davis et al. 2012 and Henry and Leone 2016) has documented the influence of tone or simple sentiment measures on immediate market reactions, narrative complexity encompasses a broader range of dimensions beyond tone alone (e.g., Muslu et al. (2015)). Additional linguistic attributes, such as specificity, optimism, directness, and aggressiveness, can also help shape investor interpretations.¹ However, because investors have limited attention time and processing capacity, complex disclosures may not be fully incorporated into prices immediately, especially when a portion of the shareholder base is distracted by other events, resulting in attenuated or staggered market reactions across short and longer horizons (e.g., Kempf et al. (2017); Basu et al. (2018); Abramova et al. (2020)).

Against this backdrop, this paper explores how institutional investor distraction, measured following Kempf et al. (2017), moderates the relation between multi-dimensional narrative content and market reactions. Consistent with theoretical models of limited attention (Hirshleifer and Teoh 2003; Hendershott et al. 2022), our premise is that, while some investors process complex disclosures quickly, distracted investors exhibit attenuated responses in the short term and continue incorporating narrative signals over an extended horizon. Rather than deferring all recognition, these investors gradually adjust their valuations as attention is reallocated and narrative complexity is fully interpreted. We propose a novel Aggregate Attribute Index (AAI) that captures diversity in linguistic attributes, including tone, optimism, specificity, directness, and aggressiveness. To isolate the incremental impact of narrative complexity, we focus on a staggered reporting setting in which firms pre-release key quantitative information

¹ In this paper, "narrative complexity" refers to the multi-faceted linguistic characteristics of a corporate disclosure that extend beyond simple sentiment or tone. It incorporates the diverse ways in which information is conveyed, accounting for additional dimensions of interest beyond tone. These additional dimensions create a nuanced narrative that requires more cognitive effort to interpret compared to more straightforward disclosures. For more detail, please see section OA-A of the online appendix.

ahead of the full 10-K filing, allowing us to better attribute price reactions to the MD&A narrative. Leveraging machine learning and natural language processing, we examine how distracted investors respond to complex disclosures over time. We find that distraction damps but does not eliminate immediate reactions in the short window (e.g., three-day window around a MD&A announcement) and leads to continued price adjustment over the longer horizon (e.g., 30-day window around an MD&A announcement). Our results suggest that disclosures convey multi-dimensional linguistic signals—not just short-run sentiment or tone—that require cognitive effort to interpret and are incorporated into prices more gradually when investor attention is initially limited.

Our focus on the attenuated and gradual reaction to narrative complexity aligns with central themes in the recent literature on disclosure and investor inattention. For instance, Bertomeu et al. (2023) show that inattention can have two opposing effects on disclosure incentives: on one hand, distracted investors respond less vigorously to new information, potentially reducing managers' willingness to make costly disclosures. On the other, they may misclassify marginal disclosures, inflating the perceived value of minimal signals and prompting some firms to disclose less than they might otherwise. When attention is extremely limited, even simple disclosures can provoke overreactions, while complex ones may be initially overlooked. However, as attention is gradually restored, investors begin to re-engage with the information, leading to more assimilation of complex narrative content. In our setting, this process manifests as a muted short-window response to narrative complexity among distracted investors, followed by continued price adjustment in the longer window. These dynamics underscore that narrative complexity is not ignored but rather processed incrementally as attention constraints relax.

Abramova et al. (2020) confirm that short-term fluctuations in institutional investors' attention—particularly among passive investors—shape managers' disclosure practices, often through additional forecasts or 8-K filings that do not necessarily enhance long-term information quality. Managers may craft simplistic or superficially positive language to allow time-constrained, distracted investors to form quick impressions while simultaneously embedding more nuanced explanations to engage nondistracted readers. Consistent with Basu et al. (2018), our findings suggest that tone alone does not significantly influence the market reactions of distracted investors. Interestingly, it is the narrative complexity embedded in the MD&A that drives meaningful price adjustments, even in the short run, with continued incorporation over longer horizons as distracted investors gradually re-engage with the disclosure.

The idea that distracted investors re-enter the market stochastically as their distraction wanes, generating “long-lived” price effects, finds its most direct parallel in the work of Hendershott et al. (2022). Their structural model shows that pricing errors can persist for weeks or months when investors fail to continuously monitor the market. These errors, driven by infrequent or distracted trading, reverse as investors eventually re-evaluate firm fundamentals. This framework supports our central hypothesis: when some investors are distracted by external events, the complexities of an MD&A are not fully priced at the time of disclosure. Leveraging the staggered release of financial information, we isolate the incremental contribution of narrative content. In periods of low attention, short-term price reactions to complex narrative attributes are damped,

and investors may initially respond only partially to these disclosures. As attention is gradually restored, pricing adjustments continue, leading to a more complete incorporation of narrative features over the subsequent period. This staggered assimilation of information is consistent with models of limited attention and supports the view that investor distraction shapes not just whether but also when narrative content affects market prices.

In the bigger picture, these findings extend the growing literature that accounts for investor distraction as a fundamental friction in financial communication. Early research on disclosure often assumed that rational investors aggregate all publicly available information immediately, but more recent studies underscore that real-world markets exhibit lags, selective reading, and capacity constraints (DellaVigna and Pollet 2009; Cohen et al. 2020). Limited attention not only shapes which firms or signals attract immediate interest but also when investors will eventually assess less salient (Hirshleifer and Teoh 2003) or more time-consuming attributes. This phenomenon is especially pertinent for textual data, where subtle linguistic cues — references to internal strategies, forward-looking statements, or risk expansions — require deeper cognitive processing. Our work shows that, while short-term returns might reflect only broad sentiment (for example, whether a filing's tone is predominantly positive or negative), the additional layers of specificity, directness, or optimism become material only after investors devote the necessary attention to parsing them. In this respect, our AAI measure captures precisely those features that distracted investors may initially overlook, making it especially well-suited to explain staggered market responses that unfold across both short and longer horizons.

From a disclosure-policy standpoint, the interplay of investor distraction and narrative complexity raises important questions. One perspective is that managers might avoid excessive detail if they suspect that inattention will undermine the immediate payoff of those disclosures. Another perspective emphasizes that, if managers predict that distracted investors will eventually reengage, they may opt to craft communications that will stand up to future scrutiny. Meanwhile, tone remains a more straightforward way to shape early impressions, yet it alone proves insufficient to fully explain the dynamics of stock returns. We observe that our multi-attribute measure better predicts price movements over an extended time window, reinforcing the argument that complexity is not merely “excess noise” in corporate communication but a vehicle for incremental information that ultimately influences valuations.

This paper makes three contributions. First, it shows that distracted investors respond differently to simplified single metrics (notably tone) than they do to multi-dimensional narratives and that these differences manifest in the timing of price reactions. Second, it extends theories of inattention, which often concentrate on how changes in investor attention correlate with broad changes in disclosure levels, by demonstrating that it is not merely the presence or absence of disclosures that matters but also their complexity. Third, it confirms the plausibility of a dynamic that research has posited but not established: sophisticated linguistic attributes in corporate filings may be only partially incorporated into prices when released during periods of high investor distraction, with their full value recognized gradually as previously distracted investors reassess the information.

These conclusions offer a cohesive explanation for the patterns observed in prior work. Institutional investors, who manage large, diversified portfolios, are often preoccupied by the most extreme performers they hold, as shown by Kempf et al. (2017). As a result, mid-performing firms issuing richly detailed MD&A disclosures may receive limited immediate attention, leading to muted market reactions to complex narrative content. Over time, as investors revisit previously overlooked filings, stock prices adjust to reflect the underlying complexity. This staggered pricing process aligns with broader evidence from the asset-pricing literature (e.g., Chen et al. (2022)), which shows that inattentive or infrequent traders contribute to persistent but ultimately reversible pricing errors.

In sum, the present analysis underscores how distracted investors, in conjunction with the multi-dimensional complexity of language, shape short- and long-horizon price dynamics. While a rudimentary focus on tone is consistent with limited attention, a more nuanced assessment of other textual attributes emerges only after distracted investors have an opportunity to fully evaluate the 10-K. This interplay has implications for understanding how quickly and by which means market participants assimilate corporate disclosures. It also provides important context for why managers often wrestle with deciding how much additional voluntary detail to provide and when. By documenting the staggered market response to our AAI in a setting characterized by temporary attention frictions, we demonstrate that narrative complexity influences how informational efficiency unfolds over time.

2 Literature review

2.1 Investor distraction

A number of theoretical models have been developed that explain how investor inattention causes security markets to reflect public information with a delay (e.g., Hong and Stein 1999; Hirshleifer and Teoh 2003; Peng and Xiong 2006; Andrei and Hasler 2015; Hendershott et al. 2022).

A growing body of empirical research suggests that investor distraction can lead to an under- or overreaction to value-relevant information. Those studies report an array of different causes for distraction, such as new products, earnings news, investor types (e.g., retail or institutional investors), analyst reports, innovative efficiency, liquidity shocks, or information about related stocks (e.g., Hirshleifer et al. 2004; Hou and Moskowitz 2005; Hong et al. 2007; DellaVigna and Pollett 2007; Barber and Odean 2008; Hirshleifer et al. 2009; Cohen and Lou 2012; Cohen et al. 2020; Cohen and Frazzini 2008; Hirshleifer et al. 2013; Bali et al. 2014; Huberman and Regev 2001; Kempf et al. 2017; Guo et al. 2020). A theme that emerges from that literature is that investors rapidly process information that is easier to understand and postpone analyzing more complex information.

In addition, several papers use investor inattention to motivate the use of heuristics that are designed to simplify problem solving (e.g., Barberis and Shleifer 2003; Peng and Xiong 2006; Hong and Stein 2007; Huang et al. 2014). That literature contemplates the possibility that investors use simple heuristics to initially assess a

financial statement but may subsequently update as they have the time to delve into its content.

As 10-Ks are typically lengthy and complex, investors need time to process them. This may lead to delayed price reactions as investors gradually unpack and absorb the information (Blankespoor et al. 2020). To illustrate this point, Cohen et al. (2020) conclude that there is “an extreme broad-based form of investor inattention to an item that is foundational to the corporate reporting process—the quarterly and annual reports—which leads to large return predictability.” They find that firms with significant changes in their 10-Ks and 10-Qs outperform nonchangers by 34 to 58 basis points per month over the *following* year. They show that the reporting changes are concentrated in the MD&A section of the reports. Their findings are intuitively compelling because the MD&A section is the one place where the SEC encourages management to “tell their own story” (Securities and Exchange Commission 2020). Cohen et al. (2020) also note: “It is not merely the difference between quantitative and qualitative information that matters for investors (as in Engelberg (2008)), but also how that qualitative information is constructed and presented.” Two studies use the Kempf et al. (2017) distraction measure to explain how the presence of distracted institutional investors may affect corporate disclosures. Abramova et al. (2020) focus on quarterly disclosure activity measured by the frequency of management forecasts and 8-K disclosures, whereas Bertomeu et al. (2023) develop a nonlinear theoretical model based on earnings forecast frequency and then estimate the effect of institutional investors’ capacity to process information on forecast frequency. We build on this research by going beyond reported disclosure frequency and examining the detailed narrative content.

2.2 Narrative content and investor reaction

Studies, such as the work of Brown and Tucker (2011), show that significant portions of a firm’s annual 10-K filings contain narratives that help contextualize financial statements. Among these, a relevant item is the MD&A. The MD&A is of particular interest because regulators encourage firms to “give the company’s perspective on the business results of the past financial year.” The literature examining investor reactions to MD&A narratives highlights the critical role that qualitative disclosures play in shaping market behavior. Studies have extensively documented how the tone and linguistic attributes of MD&A narratives influence investor decision-making and stock market outcomes. For instance, Li (2010) finds that more favorable MD&A narratives are associated with higher future earnings and stock returns, suggesting that investors view positive tone as a signal of strong future performance. Huang et al. (2014) also examine the readability of annual reports, focusing in part on MD&A disclosures, and find that less readable narratives are linked to greater information asymmetry and weaker investor responses. Their findings underscore how linguistic clarity enhances financial communication. Muslu et al. (2015) examine the role of forward-looking statements in the MD&A sections of 10-K filings. They find that, when firms’ stock prices are inefficient, they provide more forward-looking disclosures. These disclosures aim to enhance the informational efficiency of prices, although they do not fully mitigate the inefficiencies. This research underscores the strategic use of narrative

content in financial reporting to influence investors' perceptions and market behavior. It aligns with our findings that firms employ a complex mix of narrative attributes, favoring positive language when reporting better operating performance. Furthermore, Loughran and McDonald (2011) develop a specialized financial dictionary to measure tone and document that negative words in MD&A sections correlate with lower stock returns, highlighting the sensitivity of investors to language conveying risk or uncertainty.

These studies share a focus on either tone or the identification of specific phrases that reflect, for example, forward-looking statements. The mixed prior results, as highlighted by Bochkay et al. (2023), can be attributed to differences in methodologies, the attributes analyzed, and the contexts in which investor reactions are studied. For instance, while tone is often a strong explanatory variable in capturing immediate market reactions, its predictive power diminishes in settings where investors are influenced by other contextual factors, such as narrative complexity or forward-looking statements. Studies, like the work of Li (2010) and Loughran and McDonald (2011), emphasize tone and forward-looking statements as primary determinants, but their reliance on lexicon-based measures can oversimplify the richness of narrative disclosures, failing to capture subtler attributes, such as specificity or optimism. Similarly, Huang et al. (2014) and Muslu et al. (2015) show that readability and forward-looking content matter significantly in influencing investor perceptions. Yet their isolated focus does not account for interactions among these attributes. Bochkay et al. (2023) argue that these discrepancies arise because single-dimension analyses often miss the multi-dimensional nature of narrative content, which better explains investor behavior. These mixed findings underscore the need for an integrated approach that considers multiple narrative dimensions simultaneously, providing a better understanding of how investors, particularly distracted ones, process financial disclosures over time. By integrating these insights, we offer an alternative approach that investigates how distracted investors jointly process these narrative dimensions over time.

A related body of literature focuses on the methods that have been used to analyze financial narrative. Several recent papers survey the relevant research in this area (Li 2010; Loughran and McDonald 2016; Lewis and Young 2019; El-Haj et al. 2019; Bochkay et al. 2023). Instead of repeating much of that discussion here, we focus on a few papers that directly inform our analysis. We separately discuss papers that use context-specific dictionaries (Loughran and McDonald 2011) and Naïve Bayesian classification (Li 2010).²

Dictionary-based approaches rely on quantitative metrics based on word counts of only those words contained in content-specific dictionaries. Loughran and McDonald (2011) have designed a popular set of financial dictionaries that allow researchers to

² Other textual methods are frequently used in the literature either separately or in conjunction with the methods discussed here. These include readability scores (e.g., the FOG index Li (2008) and Loughran and McDonald (2014)), document length (You and Zhang 2009), keyword searches (and associated word counts) (Loughran and McDonald 2011), and cosine similarity (Lang and Stice-Lawrence 2015). The survey papers mentioned above discuss these approaches.

calculate positive and negative tone.³ Examples of papers that use context-specific dictionaries include those that have been calibrated to financial applications (Feldman et al. 2010; Loughran and McDonald 2011; Davis and Tama-Sweet 2012), while other studies employ generalist dictionaries (Muslu et al. 2015; Mayew et al. 2015).

Another approach that belongs to the family of “supervised” machine learning methods is the Naïve Bayesian classification method (Manning and Schutze 1999; Bochkay et al. 2023).⁴ A key advantage of this method, relative to dictionaries or other rule-based approaches, is that it relies on statistical inference to accommodate a more precise specification of the underlying research question.⁵ For example, Ryans (2021) uses Naïve Bayesian classification to classify firms that are associated with future write-downs and restatements.⁶

Li (2010) uses Naïve Bayesian classification to study the association between tone and financial metrics. He shows that tone is positively associated with current performance, small size, future earnings, and low growth opportunities. Our paper builds upon his work by extending the set of language attributes available for study. In Li’s own words: “Conceptually, there are at least three disclosure characteristics that are interesting to researchers: the level (how much you say), the tone (what do you mean), and the transparency (how you say it).” By expanding the number of linguistic attributes, we can more fully address “how you say it,” allowing us to determine whether other dimensions of narrative content detected in MD&As partially explain reported earnings and subsequent investor reactions.

3 Hypotheses development

We examine how firms use narrative attributes in the MD&A section and how these attributes influence investor reactions. Our analysis assumes managers possess privileged information about their firms’ operating environment, which allows them to strategically select narrative attributes to shape investor perceptions. This strategic communication, often driven by the firm’s performance, aligns with prior findings suggesting that managers use narrative disclosures not only to inform but also to influence investor behavior (Li 2010; Huang et al. 2014; Muslu et al. 2015). Extending the frameworks provided by Li (2010) and Huang et al. (2014), we hypothesize that managers are more likely to emphasize positive and optimistic attributes when their operating environment reflects strong financial performance or promising growth opportunities. Li (2010) analyzes the tone of MD&A sections and finds that more posi-

³ Loughran and McDonald (2011) also provide dictionaries to evaluate additional attributes. These include uncertainty, litigious, strong modal words, and weak modal words. The finance-specific dictionaries generally track the attributes characterized in generalist dictionaries. The difference is that dictionary-based approaches delete words with ambiguous interpretations from the more generalist word lists (e.g., Harvard General Inquirer Stone et al. (1966)).

⁴ The authors will make their Naïve Bayesian classifiers available upon request.

⁵ For instance, Naïve Bayesian classifiers can focus on specific language attributes that are not available in dictionary-based approaches.

⁶ Henry and Leone (2016) compare dictionary-based approaches to Naïve Bayesian classification and concludes that dictionary-based approaches work as well as Naïve Bayesian classification in some settings.

tive narratives are associated with higher future earnings and stock returns, suggesting that investors view positive tone as a signal of strong future performance. Similarly, Huang et al. (2014) investigate the readability of annual reports, including MD&A disclosures, and demonstrate that less readable narratives are associated with higher information asymmetry and lower market reactions, emphasizing the importance of clarity in financial communication. These studies collectively suggest that managers strategically use positive language in narrative disclosures to align with favorable operating conditions. Conversely, in periods of uncertainty or underperformance, managers may downplay negative aspects and mitigate adverse market reactions (Loughran and McDonald 2011; Bochkay et al. 2023).

Rather than focus solely on a single attribute, such as tone, we expand include additional narrative attributes, such as optimism, specificity, directness, and aggressiveness. These attributes provide complementary perspectives on how firms strategically craft their MD&A disclosures to reflect their operating environment. While tone has been widely studied and offers valuable insights, relying exclusively on it poses inherent limitations. For instance, tone alone cannot capture the depth, clarity, or forward-looking nature of a narrative, nor can it distinguish between different types of positive or negative disclosures, such as forward-looking optimism versus factual specificity (Li 2010; Muslu et al. 2015). By adopting a multi-dimensional framework, we address these shortcomings and provide a more holistic view of how firms communicate their performance and strategies. This approach recognizes the interplay between various narrative dimensions and their collective ability to convey information about a firm's operational environment. Moreover, we incorporate features of firms' operating environment, such as earnings and financial condition, to better contextualize the strategic use of narrative attributes. These elements allow us to examine whether individual narrative attributes align with firm information and how they jointly contribute to a broader understanding of financial disclosures. Given our assumption that the numbers alone are not enough, we first test whether individual narrative attributes are unrelated to economic performance. We formalize our first hypothesis in its null form as follows:

Hypothesis H1: Firms do not employ diverse linguistic attributes to contextualize reported financial performance, and the selection of narrative content is not strategically influenced by their operational environments.

We now turn our attention to the combined effect of our narrative attributes. We extend the univariate analysis proposed in Hypothesis **H1** by evaluating the extent to which firms emphasize multi-dimensional narrative attributes jointly to achieve an overarching disclosure objective. While individual narrative attributes, such as tone or forward-looking statements, have been widely studied in isolation, they are not inherently independent. Rather, managers often employ these features simultaneously to reflect a cohesive communication strategy. Following Gompers et al. (2003), who developed the Governance Index to capture systematic variation in firms' governance practices, we posit that narrative attributes exhibit similar systematic patterns when used to convey performance and strategic intent. A composite index, as argued by Gompers et al. (2003), offers a clearer interpretation of firm behavior than any single

measure. Analogously, we propose aggregating the narrative attributes into a single AAI, which reflects the common structure of how firms craft multi-dimensional narratives. The AAI for firm i at time t is defined as:

$$AAI_{i,t} = Tone_{i,t} + Optimism_{i,t} + Specificity_{i,t} + Directness_{i,t} - |Aggressiveness_{i,t}|, \quad (1)$$

where each component corresponds to the classification score derived from a dedicated Naïve Bayesian classifier.⁷

This formulation is motivated by the idea that narrative choices systematically reflect the firm's operating environment. For example, firms with strong performance may concurrently emphasize positive tone, optimism, and specificity, whereas those facing difficulties may temper these with evasive or passive language. The AAI thus serves as a scalar-valued summary of interrelated disclosure dimensions, capturing the firm's overarching communication intent. This mirrors the methodology of Gompers et al. (2003), who used a linear combination of governance metrics to uncover governance patterns. In our setting, each attribute (e.g., $Tone_{i,t}$) reflects the assessed level of a specific linguistic dimension within the MD&A. Positive scores are constructed to represent favorable disclosure characteristics. The absolute value of *Aggressiveness* is included because both aggressive and passive language typically have negative connotations. Although the AAI weights all components equally, which does not account for differential importance, Gompers et al. (2003) note that this uniform weighting ensures transparency and replicability.⁸

This formulation reflects the systematic nature of narrative disclosures, wherein each attribute contributes to the overall message conveyed by the firm. The inclusion of attributes, such as tone, optimism, specificity, and directness, adjusted by the absolute value of aggressiveness, ensures that the AAI captures both the positive and negative elements of disclosures. By aggregating these attributes, the AAI provides a better measure of narrative content than analyzing each attribute independently. Furthermore, the AAI allows us to evaluate how firms' narrative strategies align with their operating environments and how these aggregated strategies influence investor behavior. The use of AAI addresses two key limitations of single-attribute analyses. First, it accounts for the interdependence of narrative dimensions, recognizing that firms rarely employ attributes in isolation. Second, it offers a holistic measure that is easier to interpret in empirical settings, especially when examining how investors, particularly distracted

⁷ For a detailed overview of the scoring criteria for each narrative attribute, see Table OA.1 in the online appendix. To further aid interpretation, Table OA.2 provides examples of how four sample sentences are classified across the different attributes.

⁸ Assuming equal weighting across attributes affects only the computation of prior probabilities and does not constrain how the likelihood is calculated. We conduct a robustness test to assess whether our results are sensitive to alternative specifications of the AAI. We estimate $AAI_{i,t}$ and its alternative form given in Equation (10) using the first principal component of the attribute classification levels. In Table OA.7 of the online appendix, we report the results for AltAAI only, as the first principal component extracted when using all the narrative attributes captures less than 50% of cumulative variance, suggesting that it struggles in identifying a dominant latent factor. Since the reported results resemble those of our main specifications, we conclude that our proposed method is robust and insensitive to estimation mechanics.

ones, react to multi-dimensional narratives over time. In doing so, the AAI enables us to explore unified disclosure objectives and assess the systematic nature of narrative content in a way that single-attribute measures cannot. Given the above intuition, we formalize our second hypothesis in the *null* form as follows:

Hypothesis H2: Firms do not emphasize individual narrative attributes to different degrees. Aggregate narrative disclosure strategies given by AAI do not provide context to financial performance and are not predicted to be associated with the operating environment.

Hypothesis **H2** is grounded in the idea that narrative attributes in MD&A disclosures are strategically aligned with a firm's conditions. The AAI builds on this premise by aggregating these attributes to better understand a given firm's financial narratives. It posits that firms with favorable operating conditions are more likely to exhibit a systematic alignment of positive narrative attributes (e.g., tone, optimism, and specificity) in their disclosures, whereas firms with less favorable conditions may display a mix of positive and negative attributes, such as increased directness or reduced aggressiveness. The systematic nature of these narrative choices suggests that the AAI is a meaningful metric that reflects the overarching narrative strategy. By hypothesizing a relationship between the AAI and the operating environment, we aim to test whether this aggregated index can capture firm performance.

While hypotheses **H1** and **H2** focus on *whether* narrative attributes are associated with financial outcomes, our next line of inquiry examines *when* investors react to this information. Specifically, we assess whether investor distraction alters the timing of market responses to narrative content, particularly in the context of staggered reporting strategies. Building on the theory of investor distraction proposed by Kempf et al. (2017), we posit that distracted investors may only partially process the multi-dimensional content in MD&A disclosures during the initial short window following a 10-K filing. Complexity plays a central role here, as research demonstrates that more complex narratives require additional cognitive effort and time to interpret and are thus less likely to be fully absorbed right away. For instance, Li (2008) find that less readable annual reports are associated with slower investor responses and greater information asymmetry. Similarly, Huang et al. (2014) demonstrate that reduced narrative clarity can damp immediate market reactions.

Complexity in corporate disclosures arises from the multi-faceted nature of narrative attributes, which provide nuanced insights into firms' operating environments. However, processing these attributes requires investors to delve deeper when simple tonal measures do not capture the financial narrative fully. Distracted investors (Kempf et al. 2017) are less likely to allocate the necessary cognitive resources during the short ($-1, +1$) event window around the 10-K announcement date to fully assess narrative dimensions.

The theoretical model of Hirshleifer and Teoh (2003) shows that investor inattention can lead to underreaction to public information, as some investors initially overlook value-relevant signals, resulting in only partial price adjustment on the day

of release.⁹ More recently, Hendershott et al. (2022) propose a model in which some investors monitor their holdings continuously, while others engage only intermittently, depending on processing constraints. This asynchronous attention produces persistent pricing errors and predictable return reversals, as market makers bridge temporary imbalances in liquidity demand.

We hypothesize that investor distraction attenuates and staggers the market's reaction to multi-dimensional narrative content. As distracted investors eventually redirect their attention to the *focal firm*, we expect them to more fully process the richer information in the AAI, leading to continued price adjustments during the post-filing window (days +2 to +30).¹⁰ This staggered assimilation of complex narrative content shows the importance of accounting for both timing and cognitive demands when analyzing investor responses. Specifically, we posit that distracted investors partially incorporate the complexity of MD&A disclosures at the time of filing, with further adjustments occurring as attention normalizes. This behavior is consistent with Hirshleifer and Teoh (2003), who model underreaction due to limited attention, and with Hendershott et al. (2022), whose framework predicts return reversals as infrequent or distracted traders eventually reevaluate firm fundamentals. Our third hypothesis focuses on this interplay between distraction, complexity, and return timing and is stated in its *null* form as follows:

Hypothesis H3: Investor distraction does not cause or delay price reactions to multi-dimensional narrative content in MD&A disclosures.

4 Research design

4.1 Staggered reporting strategy

There is significant cross-sectional heterogeneity in firms' approaches to disclosure. Many hold earnings conference calls on the same day they file their 10-Ks. Some of these firms may simultaneously issue a press release, which often results in the filing of an 8-K. Other firms may forgo an earnings call and issue a press release instead, particularly those that do not have analyst coverage. Still others follow a *staggered* reporting strategy in which they issue a press release, possibly in conjunction with an earnings conference call, *before* they file their 10-K report. Alternatively, they may host a pre-filing earnings call without issuing a press release. Research has examined market reactions to various corporate disclosures, such as earnings announcements,

⁹ Consistent with this explanation, Hirshleifer et al. (2009) find that investors respond less to earnings announcements made on days with more unrelated news, suggesting that distraction impairs immediate market reactions. Similarly, DellaVigna and Pollett (2007) observed that earnings announcements made on Fridays — when investors are presumed less attentive — trigger weaker immediate reactions and more pronounced post-announcement drifts. These findings underscore the role of limited attention in slowing the incorporation of information into prices.

¹⁰ The term "focal firm," drawn from Kempf et al. (2017), refers to the company under study whose disclosures are evaluated for price impact in the presence of investor distraction, typically caused by competing announcements within the same investment portfolio.

press releases, and conference calls. However, these studies often treat these events in isolation without considering the diverse reporting strategies firms employ. For instance, studies, like those by Li (2010) and Huang et al. (2014), analyze the impact of MD&A disclosures and annual report readability on investor behavior, respectively, but do not account for the timing or combination of different disclosure methods. Similarly, Muslu et al. (2015) investigate forward-looking statements in MD&A sections without addressing how these narratives interact with other disclosure practices.

Focusing on staggered filers allows us to better isolate the impact of the information contained in the 10-K, particularly the MD&A section, on market reactions. Following the literature (Chapman 2018; Israeli et al. 2021), we posit that attentive investors swiftly process the information disclosed in earlier pre-release announcements (e.g., conference calls, 8-K filings, or press releases). They also rapidly incorporate any incremental information from the 10-K filing, including the narrative content (Li and Ramesh 2009; Li 2010; Loughran and McDonald 2011). In contrast, our focus is to test whether *distracted* investors assimilate the full extent of incremental MD&A disclosures more gradually, particularly when those disclosures involve complex narrative attributes that demand greater cognitive effort to interpret than simpler measures, such as overall tone.

Our empirical design explicitly recognizes the difficulty in disentangling the incremental effects of narrative content from the numbers. Since firms provide voluntary narrative to contextualize the quantitative results, quantitative and qualitative content is likely to be positively correlated. Any potential multi-collinearity makes detecting and isolating significant effects associated with narrative content challenging. To improve identification, we only include firms that *stagger* the initial reporting of financial performance and the filing of their 10-Ks. These firms typically issue press releases and host conference calls to disclose important quantitative information (e.g., earnings and revenues) prior to the 10-K filing date. By conditioning on staggered announcers, we better isolate investor reactions to the incremental information in MD&A sections. Our identification strategy does not isolate the effects of MD&A narrative because 10-Ks also include detailed quantitative information in footnotes (Amel-Zadeh and Faasse 2016). However, our focus on filing date returns for staggered announcers removes much confounding information related to key performance metrics and significantly increases our power to test the incremental impact of MD&A narrative content.¹¹

4.2 Multi-dimensional text attributes

To assess the narrative content of MD&A sections, we construct Naïve Bayesian classifiers (Hastie et al. 2009; Li 2010) for tone, optimism, specificity, directness, aggressiveness, and boilerplate language. Each attribute captures a distinct dimension

¹¹ While staggered reporting firms disclose key metrics that are important to investors at the earlier earnings announcement date, the subsequent 10-K filing will contain additional quantitative information not previously disclosed. We consider this information to still be important, particularly as it relates to the information disclosed in earnings conference calls, where analysts can ask questions about the issues they deem to be value-relevant. Hence, we propose that this information provides additional insights that attract investors' attention.

of the narrative. Tone reflects the overall sentiment of the text, distinguishing between positive and negative language, such as encouraging statements versus cautionary remarks, and aligns with traditional sentiment analysis (Loughran and McDonald 2011; Huang et al. 2014; Henry and Leone 2016). Optimism, on the other hand, captures forward-looking perspectives, identifying language that conveys either confidence or skepticism about prospects. Specificity measures the level of detail provided in the narrative, with higher scores indicating precise, fact-supported disclosures and lower scores reflecting vague or general statements. Directness assesses the clarity of the language, with higher scores denoting clear and unequivocal statements and lower ones indicating evasive or ambiguous wording. Aggressiveness evaluates the intensity of the language, distinguishing between assertive or strongly worded phrases and neutral or passive communication.

To train our Naïve Bayesian classifiers, we used a web-based survey tool where responses are crowd-sourced from 283 respondents, including members of the authors' LinkedIn networks and students enrolled in graduate accounting and finance classes.¹² Crowd-sourcing the data collection phase of our project has several advantages. First, it ensures that the results are not subject to the personal biases of a few individuals. Second, the survey design minimizes the risk of "classification fatigue" by allowing respondents to score as many sentences as they want. In addition, each respondent can connect to the survey tool with no limits or restrictions in terms of the number of accesses or session duration. These features increase our ability to collect a large amount of data in a timely and accurate manner. The Naïve Bayesian classifiers are trained to predict the narrative attributes identified by respondents in our survey. Specifically, the classifiers aim to replicate human judgment by learning from the labeled data provided by survey participants, who assessed sentences from MD&A sections across multiple narrative dimensions of tone, optimism, specificity, directness, and aggressiveness.

Respondents in our survey were presented with individual sentences from MD&A texts and asked to evaluate each sentence along these narrative attributes. For instance, respondents were asked whether a sentence conveyed a positive or negative tone, whether it was optimistic or pessimistic, specific or vague, and so on. These evaluations were provided on a five-point scale ranging from -2 (i.e., very negative, very vague) to +2 (i.e., very positive, very specific), with a score of 0 representing a neutral assessment. Additionally, respondents classified sentences as boilerplate (1 = Yes, 0 = No) when the language appeared repetitive or generic rather than firm-specific. We obtained a total of 12,192 labeled sentences which serve as the training set for the Naïve Bayesian classifiers.¹³ Using this labeled data, the classifiers are trained to predict the likelihood of a sentence being associated with a specific attribute level based on its linguistic features, such as word frequencies and context. By learning from

¹² We surveyed both academics and a variety of practitioners in financial services, auditing, consulting, asset management and financial regulation. Although it would be interesting to analyze classification differences across respondents, we did not collect demographic data because we believe that anonymity maximizes response rates. The survey remains open indefinitely, allowing the number of classified sentences to grow over time.

¹³ We provide more details on how we built the training sample in Section OA-B of the online appendix. Table OA-3 reports the definition of each narrative attributes.

these patterns, the classifiers generalize the rules of attribute identification and apply them to unlabeled sentences in the broader dataset of MD&A sections. This process ensures that the classifiers mimic human judgment with a high degree of accuracy, providing a systematic way to analyze large volumes of narrative text. By combining these attributes, we aim to capture the multi-dimensional nature of MD&A narratives, enabling a richer analysis of their content.

Overall, the Naïve Bayesian classifiers do an excellent job in predicting individual narrative attributes. The aggregate in-sample correct classification rates across all five narrative attributes ranges from 79.0% to 84.4%.¹⁴ Out-of-sample correct classification rates tend to decrease but remain quite good. A tenfold cross validation test reports aggregate correct classification rates that range from 56.3% to 84.5%.¹⁵ A detailed discussion of how we characterize multi-dimensional narrative attributes as well as the characteristics of our training set and the predicted attributes is available in the online appendix.¹⁶

4.3 Measuring investor distraction

The Kempf et al. (2017) measure of institutional investor distraction quantifies the degree of attention an institutional investor directs toward a specific firm at a particular time, assuming that institutional investors have limited monitoring capacity. The measure is based on the notion that, when an institutional investor faces an exogenous shock to an unrelated segment of that person's portfolio, attention shifts toward the affected segment and away from the unaffected focal firm. We hypothesize that this temporary diversion of attention leads institutional investors to postpone a comprehensive evaluation of the 10-K narrative content for "nonshocked" firms. Consequently, using the Kempf et al. (2017) distraction measure, we investigate whether investor distraction correlates with contemporaneous reactions to the narrative content of firm disclosures.

An attractive feature of the distraction measure is it captures exogenous variation in institutional investor monitoring intensity (attention). This exogeneity arises because the extreme economic shocks arise in other industries, which by construction are not experienced by firms that belong to the same industry as the focal firm. Kempf et al. (2017) validate their measure by empirically demonstrating that, when a focal firm has distracted investors, those investors are less likely to attend conference calls and or make annual proxy proposals. The distraction measure D is calculated following the procedure outlined by Kempf et al. (2017). First, we identify the industries that create the most investor distraction as those with the highest and lowest returns. Next, we compute how much each institutional investor is distracted based on institutional investor portfolio weightings in the extreme-return industries. Finally, we calculate the

¹⁴ The aggregate in-sample classification rates are as follows: *Tone* = 79.0%, *Optimism* = 80.7%, *Specificity* = 84.4%, *Directness* = 80.9%, and *Aggressiveness* = 81.9%.

¹⁵ The aggregate out-of-sample classification rates are as follows: *Tone* = 79.3%, *Optimism* = 83.5%, *Specificity* = 59.2%, *Directness* = 56.3%, and *Aggressiveness* = 84.5%.

¹⁶ The attribute classification frequency distributions and the related Naïve Bayes classifier accuracy rates for the training sample are reported in Table OA.4 in the online appendix. The attribute classification frequency for the full sample is reported in Table OA.5.

focal firm-level distraction score by aggregating all the distraction scores of a firm's institutional investors.

Following this outline, distraction D for firm f in quarter q is calculated as:

$$D_{fq} = \sum_{i \in f_{q-1}} \sum_{IND \neq IND_f} w_{ifq-1} \times w_{iq-1}^{IND} \times IS_q^{IND}, \quad (2)$$

where f_{q-1} denotes the set of firm f 's institutional shareholders at the end of quarter $q-1$, IND denotes a given Fama-French 12 industry, and IND_f denotes firm f 's Fama-French industry. w_{iq-1}^{IND} is the weight of industry IND in investor i 's portfolio in quarter $q-1$. IS_q^{IND} is an indicator variable that takes the value 1 if industry IND has the highest or lowest return across all 12 Fama-French industries in quarter q .¹⁷ A weighted average of the product $w_{iq-1}^{IND} \times IS_q^{IND}$ represents how much institutional investor i is distracted by unrelated industry IND .

The aggregate level of distraction for firm f is calculated as a weighted average where the weights (w_{ifq-1}) reflect two measures of investor interest: 1) the proportion of investor i 's portfolio invested in firm f and 2) the proportion of firm f 's outstanding stock held by investor i : w_{ifq-1} . w_{ifq-1} is calculated as:

$$w_{ifq-1} = \frac{QPFWeight_{ifq-1} + QPercOwn_{ifq-1}}{\sum_{i \in F_{q-1}} (QPFWeight_{ifq-1} + QPercOwn_{ifq-1})}, \quad (3)$$

where $PercOwn_{ifq-1}$ is the fraction of firm f 's shares held by investor i and $PFWeight_{ifq-1}$ is the market value weight of firm f in investor i 's portfolio. To minimize the impact of outliers and measurement error, we then sort all stocks held by investor i in quarter $q-1$ by $PFWeight_{ifq-1}$ into quintiles, denoted $QPFWeight_{ifq-1}$. Similarly, we sort firm f 's shareholders by $PercOwn_{ifq-1}$ into quintiles $QPercOwn_{ifq-1}$.

4.4 Empirical strategy

Hypothesis **H1** considers whether an association exists between narrative content, proxied by our language attributes, and the operating environment. The test relies on the following regression specification:

$$Attribute_{i,t} = \alpha + \beta \cdot \text{Controls}_{i,t} + \delta_i^{Firm} + \delta_t^{Year} + \varepsilon_{i,t}, \quad (4)$$

¹⁷ Kempf et al. (2017) motivate industry shocks as distracting events by noting that "retail investors and the media might focus 'too much' on out- and under-performers, which, in turn, might give an incentive to institutional managers to shift some of their attention to the segment most salient to their investors."

where $Attribute_{i,t}$ denotes the Naïve Bayesian classification scores for *Tone*, *Optimism*, *Specificity*, *Directness*, and the absolute value of *Aggressiveness* for firm i at time t .¹⁸

To capture firm performance, we use a vector of firm-specific control variables, denoted as **Controls** $_{i,t}$. This includes variables reflecting financial performance (ROA, free cash flow, earnings volatility), operational characteristics (number of business segments, Herfindahl Concentration Index), growth opportunities (total assets to enterprise value), changes in the operating environment (recent M&A, downsizing of operations), and financial condition (debt-to-assets ratio).¹⁹ δ_i^{Firm} denotes firm fixed effects, and δ_t^{Year} denotes year fixed effects.

We test **H2** in two ways. The first uses the OLS specification in Equation (4), and the second is a random effects ordered probit model (Crouchley 1995) as given in Equation (5) below. The benefits of the second model are twofold. The random effects model specifically addresses within-firm time series variation in its disclosure strategy. The second is that it can be used to control for the endogenous selection of attributes in our subsequent tests of investor reactions using a Heckman-type correction for selection. The ordered-probit model is estimated as:

$$AAI_{i,t} = \beta \cdot \mathbf{Controls}_{i,t} + v_i + \varepsilon_{i,t}, \quad (5)$$

where **Controls** $_{i,t}$ are the same variables described in Equation (4), v_i is the panel random effect for firm i at time t , and $\varepsilon_{i,t}$ is the stochastic disturbance term for firm i at time t . Both v_i and $\varepsilon_{i,t}$ are assumed to be independent and identically distributed as $\mathcal{N}(0, \sigma_v^2)$ and $\mathcal{N}(0, \sigma_\varepsilon^2)$, respectively.

According to this model, the probability of observing firm i 's response at time t is uniquely determined as:

$$Pr(AAI_{i,t} > k \mid \kappa, x_{i,t}, v_i) = \Phi(\beta \cdot x_{i,t} + v_i - \kappa_k), \quad (6)$$

where k denotes the cardinal values of the aggregate attribute score ($k = \{-10, \dots, +8\} \in \mathbb{Z}$), κ_k denotes the cutpoint κ at the attribute level k , and $\Phi(x_{i,t})$ is the standard normal cumulative distribution function. The cutpoints κ_k facilitate the estimation of the probability of observing an attribute score equal to k such that:

$$Pr(AAI_{i,t} = k \mid \kappa, x_{i,t}, v_i) = \Phi(\kappa_k - \beta \cdot x_{i,t} - v_i) - \Phi(\kappa_{k-1} - \beta \cdot x_{i,t} - v_i). \quad (7)$$

¹⁸ All narrative attributes are listed in Appendix A and are evaluated on a five-point scale [-2, -1, 0, +1, +2]. For example, *Tone* can be classified as positive (+2), somewhat positive (+1), neutral (0), somewhat negative (-1), and negative (-2). We use the absolute value of *Aggressiveness* because aggressive and passive narrative conveys an unfavorable message. The absolute value converts *Aggressiveness* from a five-point to a three-point scale. For more detail, see Online Appendix OA-B.

¹⁹ We use the total-assets-to-enterprise-value ratio to capture growth opportunities. Because this is the inverse of the typical market-to-book ratio, firms with more growth opportunities will have a negative association with the metric of interest. For example, if tone and growth opportunities are positively associated, in a regression model that treats tone as the dependent variable, total assets to enterprise value will have a negative coefficient.

Because $AAI_{i,t}$ is not directly observed, its location is fixed by setting the intercept in Equation (4) equal to zero.²⁰ Lastly, the Inverse Mills Ratio (IMR) for the ordered probit model of Equation (5) is calculated as follows:

$$IMR_{i,t} = \frac{\phi(\hat{\kappa}_j - \beta \cdot x_{i,t} - v_i) - \phi(\hat{\kappa}_{j+1} - \beta \cdot x_{i,t} - v_i)}{\Phi(\hat{\kappa}_j - \beta \cdot x_{i,t} - v_i) - \Phi(\hat{\kappa}_{j+1} - \beta \cdot x_{i,t} - v_i)}, \quad (8)$$

where $\hat{\kappa}_j$ is the estimated cutpoint for the attribute classification level j . For robustness purposes, we include the Inverse Mills Ratio in all our regression specifications to highlight the importance of accounting for potential endogenous narrative choice.

To test hypothesis **H3**, we estimate two regression models that use the three-day (-1,+1) and the 30-day (+2, +30) cumulative abnormal return (CAR) on the 10-K filing date as the dependent variable. Our main variables of interest are the AAI , a distraction dummy D , and their interaction term. For both specifications, we include firm-specific controls as well as industry and year fixed effects. The model is formalized as follows:

$$CAR_{i,t} = \alpha + \beta_1 \cdot AAI_{i,t} + \beta_2 \cdot [AAI_{i,t} \times D_{i,t}] + \beta_3 \cdot D_{i,t} + \gamma \cdot \text{Controls}_{i,t} + \delta_i^{Firm} + \delta_t^{Year} + \varepsilon_{i,t}, \quad (9)$$

where $CAR_{i,t}$ is either the three- or 30-day cumulative abnormal returns for firm i at time t , $D_{i,t}$ is an indicator variable that takes value 1 when the aggregate firm-level distraction measure calculated with Equation (2) (Kempf et al. 2017) exceeds sample median and 0 otherwise, $AAI_{i,t}$ is the Aggregate Attribute Index defined in Equation (1), $\text{Controls}_{i,t}$ is a vector of firm control variables that includes the tone calculated with the dictionary provided by Loughran and McDonald (2011), the FOG readability index from Li (2008), and the Inverse Mills Ratio in addition to those defined for Equation (4), δ_i^{Firm} denotes firm fixed effects, and δ_t^{Year} denotes year fixed effects.²¹

5 Empirical results

5.1 Data

Our sample period covers 2001 through 2022. The MD&A sections (Items 7 and 7A) are extracted from 10-Ks downloaded from the SEC EDGAR database and preprocessed to retain the narrative components. We obtain institutional ownership data from

²⁰ The fact that $AAI_{i,t}$ is not directly observed means that, while the AAI can be computed using the attributes we derive from the narrative analysis, it is not an exogenously given variable that can be observed or measured independently of our classification model. Instead, AAI is an aggregate measure that we construct based on the combination of various linguistic features (e.g., tone, optimism, specificity, directness, aggressiveness) scored from the MD&A sections.

²¹ We perform several robustness tests using a similar regression specification as that of Equation (9) and report the results in the online appendix. First, we exclude the COVID-19 pandemic years 2019 through 2022. Second, we regress the three-day and 30-day absolute cumulative abnormal returns. Third, we use industry-level clustered standard errors instead of firm-level ones. Results are qualitatively similar with our main specification and are reported in Tables OA.8, OA.10, and OA.12, respectively. We also report the regression results when considering the full sample of staggered and nonstaggered firms in Table OA.14.

LSEG Workspace (formerly, Eikon/Refinitiv Workspace). We also require firms to have financial statement data in Compustat and returns data in the Center for Research in Security Prices (CRSP) database. Cumulative abnormal returns are calculated using the Market Model. The market proxy is the CRSP Value-Weighted Index. We retain a total of 37,600 firm-year observations with 5,537 unique firms between 2001 and 2022. We report descriptive statistics for the variables used in the paper in Appendix Table 3, while Appendix A describes the variables used in this paper.

5.2 Descriptive statistics

We report aggregate summary statistics for all variables in Appendix Table 3. The mean AAI of 3.55, with a standard deviation of 0.97 and a median of 4.00, suggests that MD&A sections, on average, exhibit positive narrative attributes. This result aligns with research indicating that firms often craft favorable disclosures to enhance investor confidence and mitigate perceived risks (Li 2010; Loughran and McDonald 2011). The narrative content of MD&As has a slightly positive tone (mean = 0.44, SD = 0.76) and is moderately optimistic (mean = 0.50, SD = 0.57). The relatively high standard deviations of Tone and Optimism relative to their means suggest considerable variation in how firms express their views on future strategies and performance, variation that likely reflects both cross-sectional differences across firms and time-series changes within firms over time. This variability aligns with Muslu et al. (2015), who highlight the diversity in forward-looking disclosures, suggesting that firms tailor narratives to their unique operating environments. Specificity (mean = 1.66, SD = 0.56) and Directness (mean = 1.61, SD = 0.51) indicate that MD&A narratives are often detailed and straightforward in how they present information. These attributes reflect a firm's commitment to transparency and provide investors with clear insights into performance and strategy. Research, such as Davis et al. (2012), emphasizes the value of specific and readable language in reducing information asymmetry and enhancing investor comprehension. However, straightforward language at the sentence level does not preclude an overall increase in narrative complexity. The mean FOG index of 23.44, which is slightly higher than that reported in earlier studies, such as Li (2008), and suggests that the overall structure and cognitive demands of MD&A narratives have become more complex over time, driven by increased disclosure volume, regulatory expectations, and firms' use of multi-dimensional narrative strategies. This trend is consistent with the increasing sophistication of corporate disclosures and the growing emphasis on providing details to meet evolving regulatory and market expectations. The mean number of business segments (Segments = 1.68) aligns closely with prior work, such as that of Huang et al. (2014) and Muslu et al. (2015), who similarly document that firms tend to operate within a few distinct business lines. This measure often indicates operational complexity and provides context for understanding how firms communicate their performance across diverse areas of activity. Overall, our sample is consistent with prior studies, even though it is focusing on a broader period.

In Appendix Table 4, we present summary statistics comparing firms with attentive versus distracted investors. We also conduct a t-test to assess the mean differences between these two groups. This split-sample analysis allows us to investigate how

narrative attributes in MD&A disclosures as well as the underlying financial characteristics differ based on investor distraction. The results reveal distinct differences in the financial profiles of firms with distracted investors. These firms are notably more profitable, as evidenced by significantly higher return on assets (ROA) and larger free cash flows. This finding aligns with research suggesting that financially stronger firms are more likely to adopt favorable disclosure strategies to reinforce investor confidence and attract capital (Li 2010; Muslu et al. 2015). Additionally, they exhibit greater growth opportunities, as measured by a lower ratio of total assets to enterprise value (Growth). This result supports the argument that firms with high growth potential use strategic narrative disclosures to articulate their long-term vision and distinguish themselves in competitive markets (Henry 1973; Loughran and McDonald 2011).

From an operational perspective, firms with distracted investors are less likely to acquire or downsize. This stability in strategic actions may reflect an intention to maintain consistent messaging and avoid amplifying investor uncertainty, consistent with findings that managerial narratives often aim to reduce perceived risks during periods of operational change (Feldman et al. 2010; Davis et al. 2012). Furthermore, these firms operate with simpler business structures, as indicated by the presence of fewer segments. Operational simplicity has been linked to more straightforward narratives, which are easier for investors to process, particularly when attention is divided (Peng and Xiong 2006; Li 2008).

Market power and competitive positioning also differ significantly between the two groups. Firms with distracted investors exhibit higher Herfindahl index scores, reflecting greater market concentration and reduced competition. This dominant market position may afford them the flexibility to craft narratives emphasizing stability and growth without the immediate pressures of intense competition. However, these firms also demonstrate higher financial leverage. This result aligns with studies suggesting that firms with high leverage may use more detailed or optimistic narratives to counterbalance investor concerns about financial risk (Loughran and McDonald 2011; Muslu et al. 2015).

5.3 Determinants analysis for individual narrative attributes

We report the empirical results for **H1** in Appendix Table 5. These results offer insights into how firms strategically tailor their narrative attributes in financial disclosures based on financial and operational characteristics. By examining the relations between linguistic features, such as Tone (column 1), Optimism (column 2), Specificity (column 3), Directness (column 4), Aggressiveness (column 5), and firm-level variables, such as ROA and leverage, the analysis highlights how firms adjust their narrative content to align with their financial performance and strategic objectives.

Column 1 in Appendix Table 5 reports a significant positive association between the tone of management discussions and ROA, consistent with research linking positive tone to favorable financial outcomes (Henry 1973; Li 2010; Loughran and McDonald 2011; Davis et al. 2012; Huang et al. 2014). This finding supports the idea that managers strategically use positive tone to signal confidence in the firm's operational success and to emphasize strong financial performance. It also aligns with the idea

that positive tone may signal optimistic news about a firm's prospects, supporting the view that tone can predict earnings and a firm's value (Li 2010). The regression coefficient for ROA, which is large and statistically significant, aligns with the notion that firms with better performance metrics aim to maintain investor trust and attract potential stakeholders by framing their results in a positive light (Davis et al. 2012). Similarly, tone correlates positively with free cash flows, albeit with weaker statistical significance, suggesting that firms with higher liquidity tend to adopt a slightly more positive tone, potentially to signal financial stability or reassure investors about their cash management strategies. Additionally, our results show a statistically significant negative relationship with the book-to-market ratio (Growth), implying that firms with greater growth opportunities (i.e., lower book-to-market values) use more positive language in their MD&A narratives. These results indicate that tone is not applied arbitrarily but rather serves as a strategic communication tool aligned with firms' financial conditions and prospects. Firms appear to calibrate their language to both signal strength and frame their growth narrative in a favorable light. By linking tone to financial fundamentals and growth indicators, the findings underscore the strategic nature of language choices in corporate disclosures.

Moreover, a negative coefficient for the debt-to-asset ratio shows that more indebted firms tend to use a more negative tone, in line with studies like Muslu et al. (2015), suggesting that highly leveraged firms may choose a conservative tone to manage expectations and maintain credibility. Furthermore, our findings indicate that firms that are acquiring, downsizing, or using complex language in their MD&A sections adopt a more negative tone. Conversely, firms with stronger market positions (as measured by the Herfindahl index), more complex operations (more segments), or those operating in volatile industries often convey more positive messages.

The results in column 2 focus on the level of optimism in MD&A sections, revealing that firms tend to be more optimistic when they have higher growth opportunities, greater market power, or more complex operations. The finding that growth-oriented firms are more optimistic may reflect their reliance on external capital and the need to attract investor attention and support elevated valuations. However, there is a significant negative relation between optimism in narrative disclosures and a firm's debt-to-asset ratio. This indicates that more indebted firms usually adopt more conservative messaging. This finding is consistent with previous research (e.g., Li (2010); Muslu et al. (2015)), which suggests that highly leveraged firms are cautiously manage investor expectations and address potential financial risks. Interestingly, our analysis does not show a significant link between optimism and ROA or free cash flows. This challenges the assumption that current earnings strongly signal prospects, indicating instead that they more attributable to current performance.

These findings suggest that optimism in financial narratives varies and is strategically shaped by the financial health and structure of a firm. Note that optimism does not always align with a positive tone. While tone reflects the overall sentiment or positivity of the language used, optimism specifically relates to forward-looking statements and future performance expectations. This distinction, supported by prior studies (e.g., Li (2010); Loughran and McDonald (2011); Muslu et al. (2015)), highlights that firms may adopt a positive tone to frame past performance favorably while expressing caution about their prospects due to potential uncertainties or challenges.

This differentiation emphasizes the importance of analyzing multiple dimensions of corporate narratives to grasp a firm's communication strategy fully.²²

Columns 3 and 4 examine the specificity and directness of corporate narratives. The findings indicate a positive correlation between specificity and ROA, suggesting that firms with stronger performance provide more detailed and specific information. This aligns with accounting theories that advocate for disclosure transparency, as high-performing firms often disclose detailed information to signal confidence and reduce informational asymmetry between management and investors (e.g., Loughran and McDonald (2016)). Additionally, the results in column 3 reveal a positive and significant relationship between specificity and the book-to-market ratio (Growth), suggesting that firms with fewer growth opportunities are more likely to provide highly specific disclosures in their MD&A sections, whereas firms with more growth (uncertain) opportunities make less specific disclosures given the uncertainty. In contrast, for firms facing limited prospects, this may be an attempt to compensate by offering more precise and detailed explanations about their current operations or strategic plans.

The analysis of directness (column 4) supports these insights. Firms that show better operating performance and have fewer growth opportunities, as indicated by positive ROA and Growth coefficients, use clearer and more direct language in their MD&A sections. Directness in communication helps diminish investor uncertainty, potentially leading to more efficient market reactions. The accounting literature (Bloomfield 2008; Li 2010; Muslu et al. 2015) also suggests that straightforward financial narratives are linked to greater investor trust and a reduced cost of capital.

The findings on aggressiveness, measured by its absolute value and reported in column 5, reveal a negative association between leverage and acquisitions. This indicates that firms with higher leverage or those pursuing acquisitions consistently adopt less aggressive language in their MD&A narratives. This disclosure pattern is consistent with a strategic communication choice aimed at managing investor expectations during periods of financial pressure or major corporate events. By adopting a more cautious tone, these firms strive to mitigate concerns and reinforce stability. This interpretation aligns with the literature on narrative management, which shows that managers often tailor their disclosures to reduce perceived risk and preserve investor confidence during times of uncertainty (Bloomfield 2008; Li 2010; Davis and Tama-Sweet 2012; Muslu et al. 2015). While alternative explanations, such as boilerplate disclosure or regulatory compliance pressures, may contribute to the observed tone, the consistent pattern across multiple firm characteristics and narrative dimensions supports the view that these linguistic choices are part of a deliberate communication strategy.

To demonstrate the generalizability of our findings beyond the staggered reporting sample, we extend the analysis of narrative attribute determinants to the entire universe of 10-K filers with available MD&A sections between 2002 and 2022, totaling 52,882 firm-year observations. Table OA.6 reports this expanded analysis. Despite differences

²² In this context, the goal is to isolate the unique relationship between each narrative attribute and firm characteristics. Since narrative attributes, by design, are interconnected and reflect overlapping aspects of a firm's disclosure strategy, including them simultaneously in a single regression likely introduces multicollinearity, thus inflating standard errors. This not only complicates the interpretation of the results but also may obscure the unique role each attribute plays in reflecting firm characteristics.

in sample size and disclosure timing, the regression results are broadly consistent with those presented earlier for staggered announcers (Appendix Table 5).

For instance, we continue to observe a strong positive association between firm profitability (ROA) and favorable narrative attributes, particularly Tone, Specificity, and Directness. Similarly, firms with greater operational complexity and lower growth opportunities tend to craft disclosures that are more detailed and direct. The signs and significance levels of the explanatory variables remain stable across models, indicating that our AAI metric captures systematic variation in narrative strategies that extends beyond the staggered setting.

This robustness check strengthens our claim that the AAI reflects a general narrative strategy that aligns with firms' fundamentals and is not limited to any specific disclosure regime. By validating the consistency of our findings across both staggered and nonstaggered firms, we reinforce the broader applicability of AAI as a tool for textual analysis in accounting and finance research.

5.4 Determinants analysis for aggregate attribute index

The analysis presented in column 6 of Appendix Table 5 examines the combined impact of various narrative attributes—tone, optimism, specificity, directness, and aggressiveness—using the AAI as defined in Equation (1). Higher values of the AAI correspond to more favorable narrative attributes, reflecting more positive, optimistic, specific, and direct language, and less aggressiveness. The significant positive correlation between AAI and the operating environment supports **H3**, suggesting that narrative disclosures are systematically shaped by firm characteristics. Unlike analyzing individual attributes in isolation, the AAI captures the aggregate disclosure strategy, which reflects the interplay of multiple narrative dimensions. Notably, the analysis reveals a strong positive relationship between AAI and ROA and a significant negative relationship between AAI and our growth opportunities proxy, defined as the inverse of the enterprise-value-to-assets ratio (i.e., a book-to-market measure, where lower values indicate higher growth prospects). The negative coefficient on Growth implies that higher-growth firms (i.e., those with lower book-to-market ratios) score higher on the AAI. Together, these results suggest that firms with stronger profitability and better growth prospects are more likely to construct disclosures that emphasize favorable narrative attributes. This supports the view that narrative disclosures are random but are carefully crafted to reflect fundamentals and signal potential. The positive association with profitability suggests that financially sound firms may use narrative tools to reinforce credibility, while the link with growth potential indicates that expanding firms embed these prospects into their language to shape investor expectations.

In column 7 of Appendix Table 5, we extend the analysis of the economic determinants of narrative attributes using the random effects ordered probit model specified in Equation (5). This model complements the OLS regression in column 6 by explicitly accounting for the ordinal nature of the AAI and incorporating firm-level random effects to capture unobserved heterogeneity. The two models are overall aligned, thus reinforcing the conclusion that firms with superior financial performance are more likely to employ a favorable combination of narrative attributes, strategically empha-

sizing linguistic dimensions, such as tone, specificity, and directness, to achieve higher AAI scores.²³

5.5 Investor reaction to narrative content

In this section, we test hypothesis **H3**, which predicts that distracted investors (proxied by the distraction measure *D* from Kempf et al. (2017)) will exhibit a delayed reaction to the narrative content in 10-Ks. The empirical results are presented in Appendix Table 6, where we investigate this hypothesis by analyzing cumulative abnormal returns (CAR) across two time windows: a short-term reaction measured by the three-day CAR (-1,+1) (columns 1 through 4) and a longer-term reaction captured by the 30-day CAR (+2,+30) (columns 5 through 8).²⁴

Our empirical strategy includes a comprehensive set of firm controls, along with industry and year fixed effects. We also incorporate commonly used textual controls: Tone(LM), derived from the Loughran-McDonald dictionary (Loughran and McDonald 2011), and the FOG index, which captures readability (Li 2008). These controls allow us to test whether our main variable of interest, AAI, conveys incremental information beyond these standard narrative measures. In the fully specified short-window regressions (column 4), when we account for investor distraction, the interaction term $AAI \times D$ is negative and statistically significant, indicating that distracted investors do respond to the complexity captured by our aggregate index, albeit in a more attenuated and lagged fashion. Evaluated at the sample mean of AAI ($\overline{AAI} = 3.9889$), the marginal effect corresponds to a -0.56% change in CAR during the three-day window ($3.9889 \times -0.0014 \approx -0.0056$), indicating that, even in the short term, distraction impairs market response to linguistic complexity. Notably, neither Tone(LM) nor FOG predict returns in any subgroup, reinforcing the idea that traditional textual measures may be too coarse to capture the informational richness of MD&A narratives. Overall, the evidence in Appendix Table 6 shows the unique explanatory power of AAI and the importance of investor attention in determining how and when narrative complexity affects market outcomes.

In the long-window analysis (column 8 of Appendix Table 6), the results further support the explanatory power of AAI in the presence of investor distraction. For instance, the interaction term $AAI \times D$ is negative and highly significant, indicating that distracted investors continue to process and incorporate the complexity of the MD&A over time, leading to sustained price adjustments during the 30-day post-

²³ While the findings in column 7 are expected to differ slightly from those in column 6, they generally align. The observed differences in coefficient significance likely arise due to the methodological distinctions between the two approaches. For instance, the ordered probit model emphasizes the likelihood of transitions between ordinal categories of AAI, better isolating within-firm variation and mitigating potential distortions from outliers or multicollinearity.

²⁴ Table OA.14 adds nonstaggered firms to the staggered sample. Once these firms are included, the $AAI \times D$ interaction is no longer significant. The loss of significance is expected: nonstaggered firms file their 10-Ks either before or on the same day they announce earnings, hold earnings conference calls, or both, so any complexity we measure in the MD&A is likely confounded with undisclosed or concurrent quantitative information. This undermines our ability to cleanly identify the incremental effect of narrative complexity when investor attention is impaired.

filing window. This supports Hypothesis H3, which posits that attention constraints shape not just the magnitude but also the timing of narrative information assimilation. Importantly, these results point to a staggered, rather than binary, reaction pattern: distraction does not preclude an immediate response but delays the full pricing of multi-dimensional narrative content. Interestingly, the $FOG \times D$ interaction is also significant but with an opposite sign, further highlighting that traditional readability metrics may proxy for disclosure obfuscation rather than informational value. By contrast, AAI captures a broader and more strategic mix of linguistic dimensions that are cognitively demanding and thus more likely to be undervalued when attention is constrained. Evaluated at the sample mean of AAI ($\bar{AAI} = 3.9889$), the cumulative effect of narrative complexity and distraction corresponds to a -1.08% change in CAR over the 30-day post-filing window ($3.9889 \times (-0.0027) \approx -0.0108$), as reported in column 8 of Appendix Table 6. This pattern of results reinforces our central claim: narrative complexity does affect market prices, but its impact depends critically on when and whether investors are paying attention.²⁵ This behavior is consistent with research on investor inattention and gradual information incorporation, such as Hirshleifer et al. (2009) and Cohen and Lou (2012), who show that distracted investors are slower to process complex disclosures, leading to attenuated or staggered reactions to value-relevant information.

Focusing on the readability index, the positive and significant coefficient on $FOG \times D$ highlights the role of linguistic complexity in shaping investor responses under distraction. Distracted investors initially appear to penalize filings that are less readable, potentially due to the cognitive burden involved in parsing dense text. Over time, however, these investors reassess and recognize the informational content embedded in more complex narratives. Yet FOG captures only surface-level textual difficulty and does not account for the strategic use of language across multiple dimensions. The simultaneous significance of both $FOG \times D$ and $AAI \times D$ in the long-window analysis shows that AAI conveys incremental information beyond traditional readability metrics. While FOG explains part of the delayed response to textual opacity, AAI reflects a richer set of linguistic features that more fully explain the behavioral dynamics observed in distracted investor reactions. The significant negative coefficient on $AAI \times D$, even when controlling for FOG and Tone(LM), underscores AAI's robustness as a composite indicator of narrative complexity. This suggests that AAI not only complements but also meaningfully extends the explanatory power of textual measures in capturing how attention frictions shape the market's assimilation of corporate disclosures.

²⁵ In the long-window analysis (column 8), the baseline effect of investor distraction (D) corresponds to a 3.29% decline in 30-day CARs when the AAI score is zero. The interaction term $AAI \times D = -0.0027$ implies that each additional point increase in AAI further reduces CARs by approximately 27 basis points. For a firm with an AAI score of 2, this corresponds to an additional 0.54% decline, yielding a total effect of -3.83%. At the higher end of the scale, a firm with an AAI score of 4 experiences a further 1.08% reduction, resulting in a total long-window CAR of approximately -4.37%. These results underscore the central claim of the paper: the pricing of complex narrative disclosures is significantly damped when investors are distracted, and this underreaction strengthens as narrative complexity increases.

5.6 Additional analyses

In this section, we extend the analysis of hypothesis **H3** by using an alternative specification of the AAI, which excludes the tone dimension. Our objective is to investigate whether investors explicitly incorporate dimensions beyond tone when evaluating the information disclosed in MD&A sections. To achieve this, we redefine the AAI to AltAAI by excluding the tone narrative attribute, as follows:

$$AltAAI_{i,t} = Optimism_{i,t} + Specificity_{i,t} + Directness_{i,t} - |Aggressiveness_{i,t}|. \quad (10)$$

By focusing on AltAAI, we address the empirical question of whether tone itself is a significant factor in investor decision-making. Furthermore, we analyze whether both distracted and attentive investors consider tone when responding to MD&A disclosures. To test this alternative specification, we follow the regression design of Equation (9) using the AltAAI defined in Equation (10) and the same set of control variables specified in **Controls**_{*i,t*}. The model is formalized as follows:

$$\begin{aligned} CAR_{i,t} = & \alpha + \beta_1 \cdot Tone_{i,t} + \beta_2 \cdot [Tone_{i,t} \times D_{i,t}] + \\ & + \beta_3 \cdot AltAAI_{i,t} + \beta_4 \cdot [AltAAI_{i,t} \times D_{i,t}] + \\ & + \gamma \cdot Controls_{i,t} + \delta_i^{Firm} + \delta_t^{Year} + \varepsilon_{i,t}. \end{aligned} \quad (11)$$

As in our discussion of AAI, the empirical results are presented in Appendix Table 7, where we analyze CARs over two time windows: a short-term reaction measured by the three-day CAR (-1, +1) (columns 1 through 4) and a longer-term reaction captured by the 30-day CAR (+2, +30) (columns 5 through 8).²⁶

The results of Appendix Table 7 generally align with those of Appendix Table 6. For the short-window results (column 4), the interaction term AltAAI × D is negative and statistically significant, indicating that distracted investors do respond to multi-dimensional narrative complexity in the short run. Regarding sentiment, the Tone variable, derived from our Naïve Bayesian classifier, exhibits no main effect. Conversely, the interaction term Tone × D shows a negative and marginally significant coefficient. This implies that even basic sentiment cues may be less readily absorbed by distracted investors in the short window. Other standard textual features, including Tone(LM) and FOG, along with their interaction terms, remain statistically insignificant. These results underscore the distinct informational content captured by AltAAI, which appears to reflect narrative elements particularly salient for distracted investors, even in the short run, and highlight that sentiment extracted through machine learning

²⁶ We perform several robustness tests using a similar regression specification as that of Equation (11) and report the results in the online appendix. First, we exclude the COVID-19 pandemic years 2019 through 2022. Second, we regress the three-day and 30-day absolute cumulative abnormal returns. Third, we use industry-level clustered standard errors instead of firm-level ones. Results are qualitatively similar with our main specification and are reported in Tables OA.9, OA.11, and OA.13, respectively.

provides relevant information beyond what is captured by more traditional readability or dictionary-based sentiment metrics.²⁷

The longer-horizon analysis in column 8 provides clearer evidence of AltAAI's relevance for explaining post-disclosure return dynamics. The interaction term $\text{AltAAI} \times D$ is negative and statistically significant at the 10% level, indicating that distracted investors continue to process and incorporate the narrative information. At the sample mean of AltAAI ($\overline{\text{AltAAI}} = 3.5208$), the interaction effect implies an additional CAR reduction of around 119 basis points for distracted investors over the 30-day post-filing window ($3.5208 \times (-0.0034) \approx -0.0119$). When combined with the baseline effect of distraction D of -3.17%, the total return impact reaches approximately -4.37%, underscoring the economic significance of attention frictions in narrative processing. In contrast, *Tone*, derived from our machine learning classifier, remains a significant predictor of returns, but its interaction with distraction is not statistically significant. This suggests that sentiment may be more readily absorbed by all investors, while deeper narrative attributes captured by AltAAI require greater cognitive effort and are more susceptible to attention constraints. Like what we found in Appendix Table 6, $\text{FOG} \times D$ also appears significant and positive, which further shows AltAAI's incremental value beyond traditional readability and tone metrics.

In conclusion, our findings show the value of aggregate narrative indexes, such as AAI and AltAAI, as an alternative and robust measure of narrative complexity. Moreover, by excluding tone, AltAAI isolates the impact of higher order attributes, providing a more comprehensive view of how investors process multi-dimensional narrative disclosures.

6 Conclusion

This paper investigates how institutional investor distraction moderates the market's response to multi-dimensional narrative content in MD&A sections of 10-K filings. We introduce a novel measure of narrative complexity, the Aggregate Attribute Index (AAI), and its variant (AltAAI), which capture the presence and intensity of several linguistic dimensions, including tone, optimism, specificity, directness, and evasiveness. These indices are constructed using modern natural language processing and machine learning tools, enabling a nuanced and scalable assessment of corporate narrative strategy.

We show that narrative complexity is not uniformly processed by all investors: the market reaction depends critically on the attentiveness of institutional investors. In particular, we find that the interaction between distraction and narrative complexity, measured through AAI and AltAAI, is statistically significant in both the short- and long-window regressions, but its economic effect strengthens over the longer horizon. For instance, in the 30-day post-filing window, a one-unit increase in AAI corresponds

²⁷ We also report the results of a similar regression design that considers the full sample of staggered and nonstaggered firms in Table OA.15. In contrast to the staggered sample, tone in the full sample regression only exhibits mild statistical significance, likely because its sentiment signal is easier to extract and remains salient, even when narrative and quantitative disclosures are confounded, unlike higher-order attributes captured by AltAAI.

to a roughly 27 basis-point reduction in CAR among distracted investors. This supports the notion that complexity induces slower processing, especially when attention is limited. Importantly, we observe that tone alone does not always significantly influence the investment decisions of these investors, which further suggests the value of incorporating a broader set of narrative attributes into the analysis. These attributes help shape investor reactions and pricing strategies, underscoring the limitations of relying on unidimensional textual metrics.

These findings challenge the assumption that securities markets always incorporate value-relevant information instantaneously, identifying two interacting frictions, narrative complexity and temporary investor distraction, that contribute to delayed market responses. Firms should carefully consider how these factors shape investor interpretation of complex narratives, tailoring their communication strategies to enhance clarity and ensure timely engagement with their disclosures.

In addition, this study highlights the transformative role of machine learning and natural language processing in uncovering the intricacies of disclosure narratives. These advanced techniques enable a deeper analysis of how firms communicate through their MD&A sections, providing new insights into how different types of investors, particularly distracted ones, respond to complex narrative content.

A Variable definitions

Table 1 List of the variables used in the paper and their definitions

Variable	Description
CAR three-day	three-day (-1, +1) Cumulative abnormal return based on the market model. Captures short-term abnormal returns around an event.
CAR 30-day	30-day (+2, +30) Cumulative abnormal return based on the market model. Captures longer-term abnormal returns to assess delayed investor reactions.
AAI	Aggregate Attribute Index, defined as the sum of the Tone, Optimism, Specificity, and Directness scores minus the absolute value of Aggressiveness (see Equation (1)).
AltAAI	Alternate Aggregate Attribute Index, defined as the sum of the Optimism, Specificity, and Directness scores minus the absolute value of Aggressiveness (see Equation (10)).
Tone	Naïve Bayes classification score for Tone based on the document-level classification of the MD&A for firm i in year t . Reflects the sentiment of the narrative.
Optimism	Naïve Bayes classification score for Optimism based on the document-level classification of the MD&A for firm i in year t . Reflects forward-looking positivity.
Specificity	Naïve Bayes classification score for Specificity based on the document-level classification of the MD&A for firm i in year t . Measures the level of detail and precision in the narrative.
Evasiveness	Naïve Bayes classification score for Evasiveness based on the document-level classification of the MD&A for firm i in year t . Measures how indirect or vague the narrative is.

Table 1 continued

Variable	Description
Aggressiveness	The absolute value of the Aggressiveness score, indicating the level of assertive or forceful language in the MD&A.
Tone(LM)	The Loughran-McDonald tone metric calculated using their word list (Loughran and McDonald 2011).
FOG	The FOG readability index as in Li (2008).
D	Dummy variable indicating distracted investors that takes a value of 1 if distraction is above the sample median and 0 otherwise.
ROA	Return on assets, calculated as net income divided by total assets. Reflects the firm's profitability.
Free cash flow	Free cash flow is calculated as the ratio of free cash flow to lagged total assets where free cash flow is defined as earnings before interest, taxes, depreciation and amortization (EBITDA) minus capital expenditures.
Growth	Total assets to enterprise-value ratio, a proxy for growth opportunities, where higher values suggest fewer opportunities.
Leverage	Debt-to-assets ratio, calculated as long-term debt divided by total assets. Measures the firm's leverage.
Acquire	Dummy variable indicating a significant increase in firm size based on total assets. Takes a value of 1 if total assets increased by more than 33% from the prior year and 0 otherwise.
Downsize	Dummy variable indicating a significant decrease in firm size based on total assets. Takes a value of 1 if total assets decreased by more than 33% from the prior year and 0 otherwise.
Herfindahl Index	Herfindahl index for firm i at year t is calculated as the ratio of revenues for firm i relative to industry average revenues based on the firm i 's two-digit SIC.
Segments	Sum of Compustat segment identifiers (SID) as $\log(1 + \text{Number of segments})$.
Earnings Volatility	Industry-adjusted earnings volatility, calculated as the standard deviation of ROA over the past five years. Indicates the stability of earnings relative to industry peers.
Complexity	The natural logarithm of the number of words divided by the number of sentences in the MD&A of firm i at year t .
IMR	Inverse Mills Ratio, used to control for potential selection bias in the analysis.

B Tables

Table 2 Sample Selection

Initial sample of MD&A sections	163,567
Delete MD&A sections containing fewer than 10 sentences	(19,767)
Delete observations that reflect a 10-K refiling	(1,040)
	142,760
Delete observations not in Compustat or Calbench	(58,184)
	84,576
Delete observations not in CRSP	(13,097)
	71,479
Delete nonstaggered announcers	(29,486)
	41,993
Delete observations if 13-F filing is unavailable or missing FOG	(4,393)
Final sample	37,600

This table reports the sample selection process

Table 3 Summary Statistics

	Mean	SD	p25	p50	p75
CAR 3-day	-0.0008	0.0694	-0.0219	-0.0010	0.0199
CAR 30-day	-0.0020	0.1764	-0.0779	-0.0027	0.0691
AAI	3.9889	1.4024	3.0000	4.0000	5.0000
AltAAI	3.5518	1.1147	3.0000	4.0000	4.0000
Tone	0.4371	0.7561	0.0000	0.0000	1.0000
Optimism	0.5019	0.5669	0.0000	0.0000	1.0000
Specificity	1.6593	0.5629	1.0000	2.0000	2.0000
Directness	1.6090	0.5097	1.0000	2.0000	2.0000
Aggressiveness	0.2183	0.5422	0.0000	0.0000	0.0000
Tone(LM)	-0.3488	0.1618	-0.4421	-0.3559	-0.2554
FOG	23.4451	2.7326	20.8655	23.9936	25.5759
D	0.3467	0.4759	0.0000	0.0000	1.0000
ROA	-0.0114	0.1772	-0.0110	0.0208	0.0708
Free Cash Flow	0.0197	0.2783	0.0112	0.0622	0.1395
Growth	0.6462	0.3285	0.3897	0.6642	0.9338
Leverage	0.1277	0.1789	0.0000	0.0109	0.2360
Acquire	0.1399	0.3468	0.0000	0.0000	0.0000
Downsize	0.0259	0.1587	0.0000	0.0000	0.0000
Herfindahl Index	0.0220	0.0750	0.0001	0.0009	0.0080
Segments	1.6835	0.6550	1.3863	1.3863	2.3026

Table 3 continued

	Mean	SD	p25	p50	p75
Earnings Volatility	0.1755	0.1079	0.0795	0.1792	0.2551
Complexity	3.4852	0.1126	3.4096	3.4778	3.5536
IMR	-0.0254	1.7059	0.2681	0.2906	0.3977

Summary statistics for the full sample of 37,600 staggered 10-K filings. The table reports the number of observations (N), the mean (Mean), standard deviation (SD), 25th percentile (p25), Median (p50), 75th percentile (p75)

Table 4 Univariate Differences - Firms with Distracted vs. Non-Distracted Investors

	Non-distracted Investors		Distracted Investors		Mean Diff.	t-stat
	D = 0		D = 1			
	Mean	SD	Mean	SD		
CAR 3-day	-0.0019	0.0712	0.0013	0.0660	-0.0032	-4.2***
CAR 30-day	-0.0055	0.1852	0.0045	0.1584	-0.01	-5.26***
AAI	3.9070	1.4436	4.1433	1.3073	-0.2362	-15.59***
AltAAI	3.5208	1.1374	3.6103	1.0681	-0.0895	-7.41***
Tone	0.3862	0.7733	0.5330	0.7127	-0.1467	-17.98***
Optimism	0.4820	0.5713	0.5394	0.5566	-0.0574	-9.36***
Specificity	1.6553	0.5896	1.6668	0.5087	-0.0115	-1.88*
Evasiveness	1.6154	0.5095	1.5969	0.5100	0.0186	3.36***
Aggressiveness	0.2319	0.5615	0.1927	0.5027	0.0392	6.67***
Tone(LM)	-0.3471	0.1571	-0.3519	0.1702	0.0048	2.75***
FOG	23.2023	2.7068	23.9025	2.7226	-0.7001	-23.82***
ROA	-0.0277	0.1948	0.0193	0.1329	-0.047	-24.65***
Free Cash Flow	0.0003	0.2977	0.0562	0.2331	-0.0559	-18.61***
Growth	0.6655	0.3290	0.6099	0.3245	0.0556	15.66***
Leverage	0.1083	0.1713	0.1643	0.1871	-0.0559	-29.17***
Acquire	0.1538	0.3607	0.1136	0.3174	0.0401	10.7***
Downsize	0.0317	0.1752	0.0149	0.1211	0.0168	9.78***
Herfindahl Index	0.0182	0.0670	0.0290	0.0877	-0.0108	-13.26***
Segments	1.6604	0.6360	1.7271	0.6874	-0.0667	-9.4***
Earnings Volatility	0.1740	0.1085	0.1784	0.1067	-0.0044	-3.77***
Complexity	3.4814	0.1128	3.4923	0.1120	-0.0109	-8.94***
IMR	-0.0140	1.7411	-0.0470	1.6375	0.033	1.79*

Panel B: Summary statistics and univariate differences for firms with non-distracted (D = 0) and distracted (D = 1) investors, based on 24,565 and 13,035 staggered 10-K filings, respectively. The table reports the number of observations (N), the mean (Mean), standard deviation (SD), the mean differences (Mean Diff) as well as the t-stat associated to the t-test. ***, **, and * respectively denote significance at the 1%, 5%, and 10% levels

Table 5 Economic Determinants of MD&A Narrative Attributes and Strategic Attribute Selection

	(1) Tone	(2) Optimism	(3) Specificity	(4) Directness	(5) Aggressiveness	(6) AAI	(7) AAI
ROA	0.4680*** (11.30)	-0.0042 (-0.13)	0.1425*** (3.52)	0.1516*** (4.32)	-0.0123 (-0.34)	0.7702*** (8.51)	0.6639*** (10.67)
Free Cash Flow	0.0364* (1.70)	0.0133 (0.72)	0.0308 (1.40)	0.0258 (1.07)	0.0003 (0.01)	0.1061** (2.02)	0.1091*** (2.64)
Growth	-0.1765*** (-8.15)	-0.2560*** (-14.90)	0.0457*** (2.61)	0.0588*** (3.67)	-0.0649*** (-3.70)	-0.2631*** (-6.16)	-0.1760*** (-6.77)
Leverage	-0.1970*** (-4.44)	-0.2816*** (-7.92)	0.1151*** (3.31)	0.0702** (2.05)	-0.2256*** (-6.17)	-0.0678 (-0.82)	0.0147 (0.31)
Acquire	-0.1057*** (-8.68)	-0.0490*** (-5.60)	0.0144 (1.46)	0.0150* (1.82)	-0.0135 (-1.40)	-0.1118*** (-4.78)	-0.0810*** (-4.43)
Downsize	-0.0573** (-1.96)	-0.0309* (-1.69)	-0.0235 (-0.97)	-0.0071 (-0.39)	-0.0064 (-0.31)	-0.1124** (-2.18)	-0.0415 (-1.02)
Herfindahl Index	0.6065*** (5.02)	0.3051*** (3.44)	-0.2209*** (-2.62)	-0.2382*** (-3.37)	0.0156 (0.17)	0.4370** (2.28)	0.9503*** (6.74)
Segments	0.0961*** (7.77)	0.0213** (2.41)	0.0051 (0.54)	-0.0161* (-1.79)	0.0118 (1.23)	0.0947*** (3.88)	0.0803*** (5.42)
Earnings Volatility	0.2417* (1.91)	0.1051 (1.17)	0.0662 (0.65)	-0.1140 (-1.41)	-0.0280 (-0.29)	0.3270 (1.41)	0.0615 (0.54)
Complexity	-0.3581*** (-5.30)	-0.2287*** (-4.44)	0.3600*** (5.92)	0.2880*** (5.35)	-0.0995* (-1.86)	0.1608 (1.08)	0.5891*** (7.22)
Constant	1.6275***	1.4466***	0.3444	0.6101***	0.6224***	3.4062***	

Table 5 continued

	(1) Tone	(2) Optimism	(3) Specificity	(4) Directness	(5) Aggressiveness	(6) AAI	(7) AAI
Model	(6.79) OLS	(7.91) OLS	(1.59) OLS	(3.21) OLS	(3.34) OLS	(6.48) OLS	Ordered Probit
Observations	37,600	37,600	37,600	37,600	37,600	37,600	37,600
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Firm RE	No	No	No	No	No	No	Yes
Adj. R ²	0.168	0.269	0.0982	0.137	0.0812	0.126	-

The table reports panel regressions of attribute classification levels for our sample of firms that have MD&A section in the 10-K reports from 2002 to 2022 (columns 1 through 6) and coefficient estimates for a random effects ordered probit regression of the aggregate attribute score, as given in Equation (5) (column 7). The unit of observation is the firm-year. The dependent variables are the attribute classification levels predicted by the Naïve Bayes classifiers (Tone, Optimism, Specificity, Directness) and the Aggregate Attribute Index (AAI), which are reported in separate columns. We transform the Aggressiveness classification by multiplying the absolute value of the Aggressiveness attribute by -1 so that a negative value can be interpreted as an unfavorable value. This is analogous to how Aggressiveness is used to calculate AAI. All OLS regression models include firm and year fixed effects, while the ordered probit include random effects. Standard errors are clustered by firm. ***, **, and * respectively denote significance at the 1%, 5%, and 10% levels

Table 6 Market Reaction Regression to AAI

	(1) CAR (-1, +1)	(2)	(3)	(4)	(5) CAR (+2, +30)	(6)	(7)	(8)
AAI $\times$ D	-0.0014*** (-2.65)	-0.0014*** (-2.69)	-0.0014*** (-2.72)	-0.0014*** (-2.69)	-0.0042*** (-2.96)	-0.0038*** (-2.69)	-0.0036*** (-2.57)	-0.0027* (-1.92)
Tone(LM) $\times$ D	-0.0031 (-0.88)	-0.0030 (-0.86)	0.0005 (0.13)	0.0001 (0.02)	0.0039 (0.41)	-0.0007 (-0.08)	0.0001 (0.01)	-0.0015 (-0.15)
FOG $\times$ D	0.0002 (0.58)	0.0002 (0.61)	-0.0000 (-0.03)	-0.0000 (-0.04)	-0.0022*** (-3.15)	-0.0013* (-1.94)	0.0020** (2.43)	0.0019*** (2.30)
AAI	0.0010*** (3.23)	0.0006* (1.79)	0.0009*** (2.85)	0.0006 (1.59)	0.0032*** (3.53)	0.0038*** (3.48)	0.0016* (1.69)	0.0026** (2.38)
Tone(LM)	-0.0010 (-0.39)	-0.0021 (-0.86)	-0.0033 (-1.14)	-0.0042 (-1.45)	0.0065 (1.02)	0.0136** (2.09)	-0.0061 (-0.83)	0.0064 (0.85)
FOG	-0.0000 (-0.07)	-0.0000 (-0.19)	0.0003 (0.91)	0.0001 (0.49)	-0.0026*** (-6.32)	-0.0024*** (-5.41)	-0.0002 (-0.27)	-0.0001 (-0.11)
D	0.0035 (0.43)	0.0030 (0.37)	0.0095 (0.87)	0.0088 (0.81)	0.0815*** (4.20)	0.0618*** (3.20)	-0.0364 (-1.54)	-0.0329 (-1.40)
ROA		0.0160*** (4.03)		0.0168*** (4.13)		-0.0219** (-2.03)		-0.0252** (-2.30)
Free Cash Flow		-0.0008 (-0.49)		-0.0014 (-0.79)		0.0182*** (3.83)		0.0155*** (3.32)
Growth		0.0043*** (3.30)		0.0035** (2.52)		0.0644*** (17.05)		0.0677*** (16.57)
Leverage		0.0036 (1.51)		0.0054* (1.87)		-0.0043 (-0.69)		0.0046 (0.66)

Table 6 continued

	(1) CAR (-1, +1)	(2)	(3)	(4)	(5) CAR (+2, +30)	(6)	(7)	(8)
Acquire		0.0009 (0.84)		0.0006 (0.55)		-0.0073** (-2.51)		-0.0080*** (-2.78)
Downsize		0.0075 (1.33)		0.0070 (1.24)		0.0156 (1.39)		0.0089 (0.80)
Herfindahl Index		-0.0031 (-0.80)		-0.0032 (-0.57)		-0.0103 (-0.99)		-0.0330** (-2.49)
Segments		0.0006 (1.23)		0.0007 (1.24)		-0.0015 (-1.15)		-0.0060*** (-4.54)
Earnings Volatility		0.0136*** (3.53)		0.0135 (1.49)		0.1003*** (10.33)		0.0832*** (3.20)
Complexity		-0.0040 (-1.20)		-0.0026 (-0.74)		-0.0026 (-0.28)		0.0006 (0.07)
IMR		-0.0002 (-1.07)		-0.0002 (-1.14)		0.0003 (0.46)		0.0005 (0.79)
Constant	-0.0057 (-1.36)	0.0032 (0.27)	-0.0126* (-1.80)	-0.0061 (-0.46)	0.0451*** (3.86)	-0.0072 (-0.22)	-0.0047 (-0.25)	-0.0589* (-1.71)
Observations	37,600	37,600	37,600	37,600	37,600	37,600	37,600	37,600
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Industry FE	No	No	Yes	Yes	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	No	No	Yes	Yes
Adj. R2	0.001	0.002	0.004	0.005	0.004	0.019	0.037	0.049

Regression results for three-day and 30-day market returns using AAI as main variable. The table reports t-statistics in parentheses using heteroskedasticity-robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels (two tailed), respectively

Table 7 Market Reaction Regression to AltAAI

	(1) CAR (-1, +1)	(2)	(3)	(4)	(5) CAR (+2, +30)	(6)	(7)	(8)
AltAAI × D	-0.0012* (-1.96)	-0.0012* (-1.91)	-0.0013** (-2.01)	-0.0012* (-1.94)	-0.0046** (-2.53)	-0.0040** (-2.21)	-0.0043** (-2.44)	-0.0034* (-1.90)
Tone × D	-0.0016 (-1.61)	-0.0017* (-1.75)	-0.0017* (-1.68)	-0.0018* (-1.76)	-0.0033 (-1.25)	-0.0031 (-1.18)	-0.0019 (-0.74)	-0.0008 (-0.32)
Tone(LM) × D	-0.0029 (-0.84)	-0.0028 (-0.80)	0.0006 (0.17)	0.0003 (0.07)	0.0034 (0.36)	-0.0011 (-0.12)	-0.0009 (-0.09)	-0.0025 (-0.25)
FOG × D	0.0001 (0.52)	0.0002 (0.58)	-0.0000 (-0.05)	-0.0000 (-0.06)	-0.0022*** (-3.19)	-0.0014** (-1.98)	0.0020** (2.43)	0.0019*** (2.29)
AltAAI	0.0006 (1.56)	0.0004 (0.98)	0.0006 (1.51)	0.0004 (0.88)	0.0026** (2.15)	0.0031** (2.32)	0.0015 (1.23)	0.0019 (1.44)
Tone	0.0016*** (2.76)	0.0010 (1.62)	0.0014** (2.18)	0.0009 (1.38)	0.0045** (2.55)	0.0055*** (2.78)	0.0017 (0.91)	0.0043** (2.17)
Tone(LM)	-0.0013 (-0.54)	-0.0023 (-0.93)	-0.0035 (-1.21)	-0.0043 (-1.49)	0.0058 (0.90)	0.0131** (1.99)	-0.0061 (-0.81)	0.0061 (0.81)
FOG	0.0000 (0.04)	-0.0000 (-0.16)	0.0003 (0.91)	0.0001 (0.49)	-0.0026*** (-6.20)	-0.0024*** (-5.37)	-0.0002 (-0.27)	-0.0001 (-0.13)
D	0.0036 (0.44)	0.0028 (0.34)	0.0092 (0.86)	0.0086 (0.79)	0.0828*** (4.25)	0.0625*** (3.23)	-0.0353 (-1.49)	-0.0317 (-1.35)
Observations	37,600	37,600	37,600	37,600	37,600	37,600	37,600	37,600

Table 7 continued

	(1) CAR (−1, +1)	(2)	(3)	(4)	(5) CAR (+2, +30)	(6)	(7)	(8)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Industry FE	No	No	Yes	Yes	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	No	No	Yes	Yes
Adj. R2	0.001	0.002	0.004	0.005	0.004	0.019	0.037	0.050

Regression results for three-day and 30-day market returns using AltAAI as defined in Equation (10) as main variable (AltAAI). Tone is reported as an additional control. The table reports t-statistics in parentheses in parentheses using heteroskedasticity-robust standard errors clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels (two tailed), respectively

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